



NITI Aayog

DRAFT



SECTOR AT A GLANCE

Spatial Planning, Design and Development

Urban Sector in India

2023-Urban-SAG-04

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Spatial Planning, Design and Development

Urban Sector in India

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Introduction

This document serves as a primer for understanding **spatial planning, design and development** in urban India. This document provides basic information on the sector – the fundamental concepts, key trends, central government priorities and example of potential interventions.

This document attempts to answer the following questions:

- What is spatial planning?
- What is urban design?
- What mechanisms are used to implement spatial plans?
- What is the status of city planning in India?
- How does urban land management relate to ease of doing business?
- How is urban development governed?
- What are central government priorities?
- Relevant guidelines
- What are potential areas of intervention going forward?

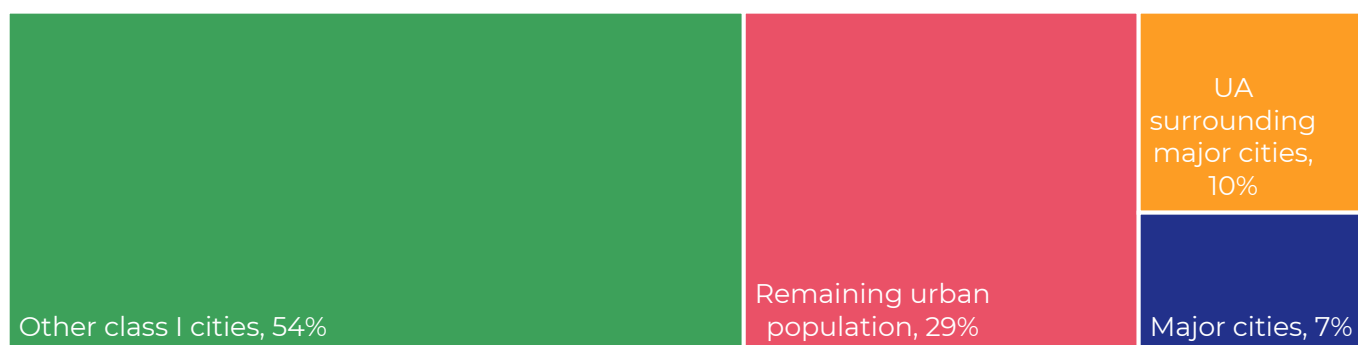
What is spatial planning and urban design, and their significance for Viksit Bharat 2047?

A. India's future urban growth expected to come from smaller towns and outgrowths

As of 2023, the share of urban population in India is expected to be more than 35%, which denotes that there is untapped potential for further urbanisation in the years to come. By the year 2047, more than half of India's population will be categorised as 'urban'.¹

Past trends of urban growth indicates that *smaller towns and agglomeration areas shall be the major contributors to future urban growth* as well. Between 2001 and 2011, the Class I cities (population greater than one lakh) which lie outside the urban agglomerations contributed to more than half of the urban population growth. Urban agglomerations displayed a decadal growth rate of more than 30% whereas Tier-1 cities grew at a decadal rate of 14.5%, between 2001 and 2011.

Figure 1: Contributors to urban population growth between 2001 to 2011 (2011) (growth share in %)

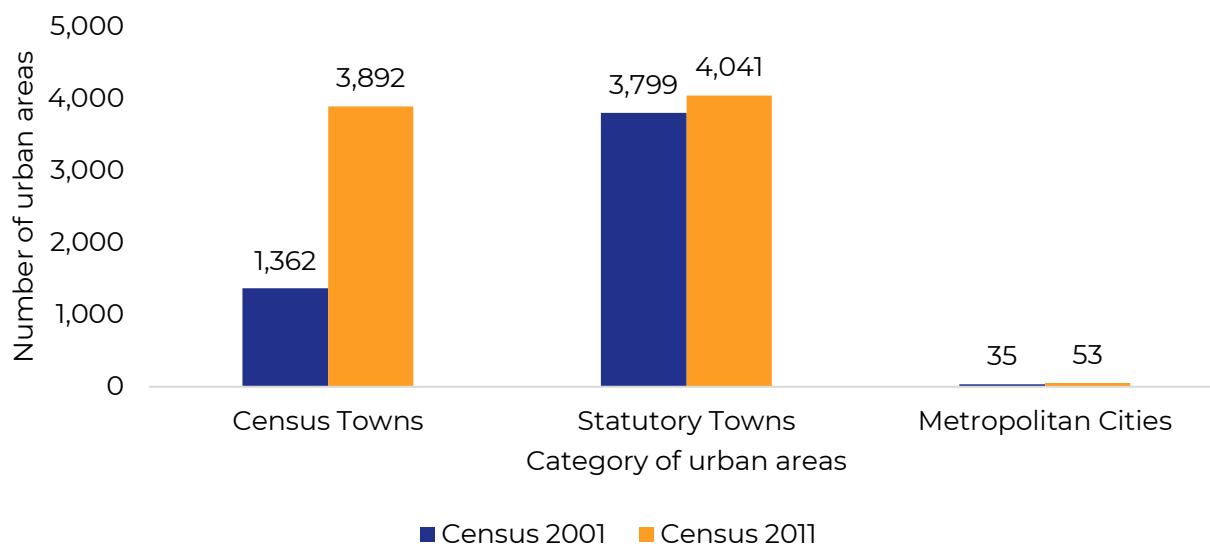


Source: Census of India 2011²

¹ UN-Habitat. (2018). *World Urbanization Prospects*. World Bank.

² Office of Registrar General & Census Commissioner, India. (2011). *Census of India*. New Delhi

Figure 2: Increase in 'urban' primarily came from census towns and statutory towns (2011)



Source: NITI Aayog, 2021³

B. Spatial planning crucial for managing India's future urban growth

What is the definition of spatial planning?

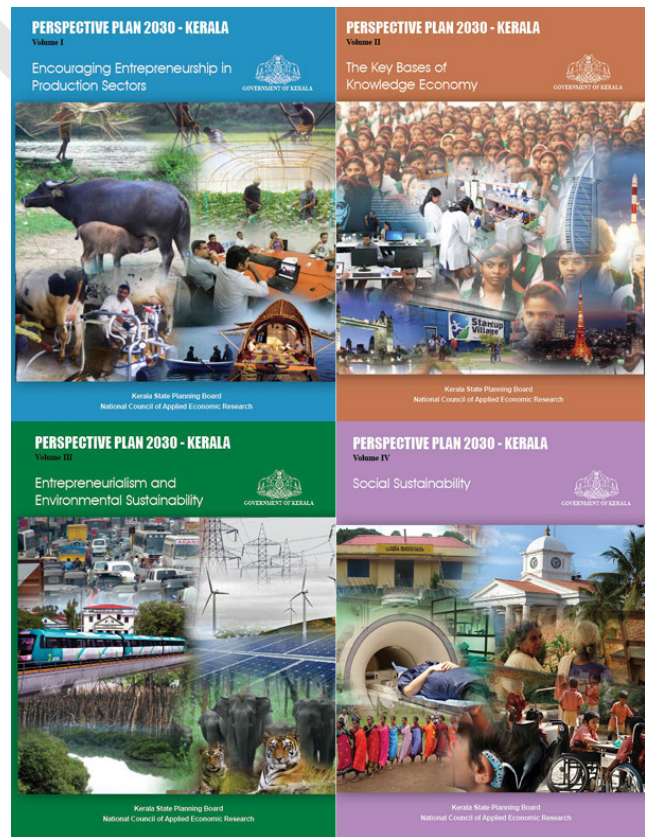
Spatial planning in context of urban areas, also denoted as urban planning/ town planning, refers to the city or region plans built upon the strategy developed by the public sector (city/ state government) to influence the distribution of people and activities in areas of different urban scales – taking (ideally) into consideration the various geographical, environmental, demographic and socio-economic features of the urban area. The outputs of the process of spatial planning typically encompasses proposed land use, proposed basic services and infrastructure, and proposed mobility plans. The city plans, referred as development plans or masterplans, are statutory in nature.

What are the different levels of urban planning generally observed in India?

As per the Urban and Regional Development Plans Formulation and Implementation (URDPFI) Guidelines, the spatial planning framework in general suggests preparation of five levels of interdependent plans, from the state-level to micro level (local level). These include both non-spatial and spatial plans. However, not all States/ cities follow the various hierarchies of planning. The following table provides an overview of type of plans and their scales with some examples:

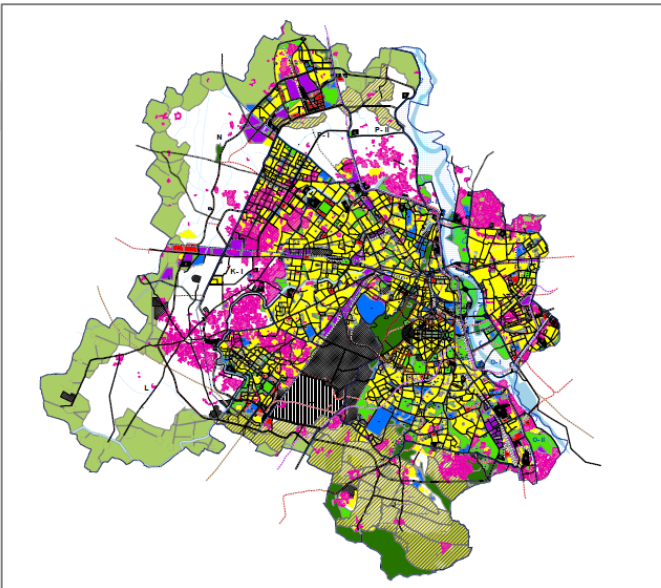
³ NITI Aayog. (2021). *Reforms in Urban Planning Capacity in India*. New Delhi: NITI Aayog.

Table 1: Levels of spatial planning in urban India, as per URDPFI guidelines

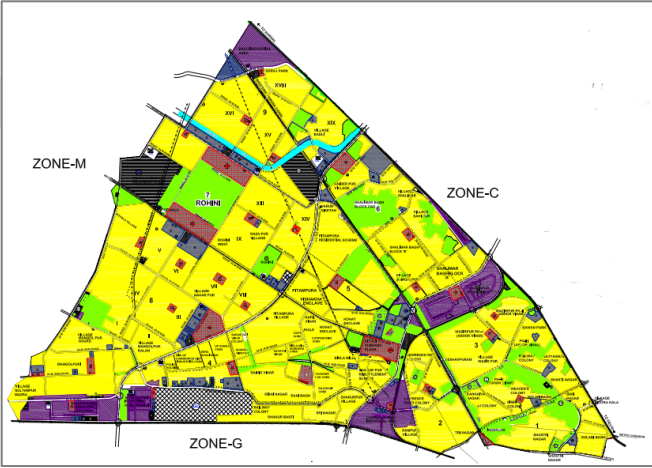
Level of planning	Scale (as per the URDPFI guidelines)	Typical contents of the plan (not an exhaustive or prescriptive list)	Function(s) of the plan	Representative image
Perspective Plan (Strategic, non-statutory plan) Typical frequency of preparation: 20-30 years Prepared by the states' Urban Development Department or the Town and Country Planning (T&CP) department e.g., Kerala State Planning Board Approving authority: State Government	1:2,50,000 to 1:1,00,000 (for larger regions) 1:1,00,000 to 1:50,000 (for smaller regions)	• Social, economic and spatial development goals, policies and priorities relating to all those urban activities that have spatial implications • Long-term policies regarding development of infrastructure and resource mobilisation	A long-term framework providing the state government the goals, policies, strategies and general programmes of the urban local authority regarding spatio-economic development of the settlement under its governance⁴	 e.g., Kerala Perspective Plan 2030⁵

⁴ Rajiv Gandhi National Institute of Youth Development. (2020). *Plan - Perspective Plan - Purpose Of Perspective Plan Preparation*.

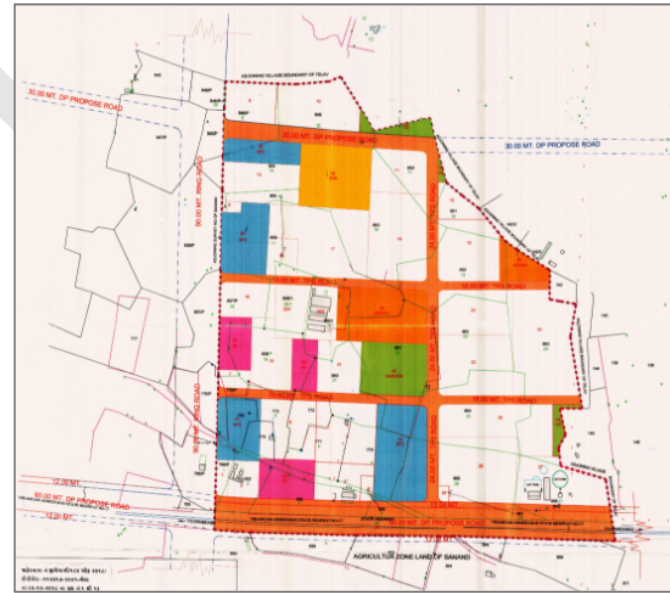
⁵ Kerala Planning Board. (2015, January 1). *Economic Review 2015 - Chapter 9: Perspective Plan 2030*. Kerala State Planning Board.

Level of planning	Scale (as per the URDPFI guidelines)	Typical contents of the plan (not an exhaustive or prescriptive list)	Function(s) of the plan	Representative image
Development Plan or Masterplan (Statutory spatial plan) Typical frequency of preparation: 10-30 years, with provision for review every five years Prepared by the city planning authority or the Urban Local body e.g., AUDA Approving authority: Municipal Corporation/ Development Authority/ Metropolitan Planning Committee	1:10,000 – 1:8,000 (as per State Provision)	• Proposals of zoning for various purposes • Designation of land for public purposes • Transport and Communication • Water supply, drainage etc. • Sites for community facilities • Sites for industries • Preservation, conservation proposals of natural sceneries • Preservation of features of historical, natural, architectural, heritage interest • Proposals for flood control etc. • Development Control & Promotion Regulations	To guide and regulate the development of cities, managing urbanisation as well as spatial sustainability	 e.g., Masterplan for Delhi⁷

⁷ Delhi Development Authority. (2021). *Draft Master Plan for Delhi - 2041*. New Delhi.

Level of planning	Scale (as per the URDPFI guidelines)	Typical contents of the plan (not an exhaustive or prescriptive list)	Function(s) of the plan	Representative image
Zonal Development Plan (Statutory spatial plan) Typical frequency of preparation: 10 years Prepared by the city planning authority or the Urban Local body e.g., Delhi Development Authority Approving authority: Municipal Corporation/ Development Authority/ Metropolitan Planning Committee	1:2,000 (as per ZDP Delhi) to 1:14,000 (as per ZDP Jaipur)	• Land use plan confirming to the masterplan • Location and extent of land uses • More detailed circulation pattern • Special objectives of the zone if any • Allocation of use zones into further use premises	A detailed land use plan prepared at a sub-city zonal level outlining the development strategies at a local level. It acts as a link between the masterplan and the layout plans	 e.g., Zonal Development Plan for Delhi⁸

⁸ Delhi Development Authority. (2021). *Zonal Development Plan (as per MPD-2021): Zone H, New Delhi*.

Level of planning	Scale (as per the URDPFI guidelines)	Typical contents of the plan (not an exhaustive or prescriptive list)	Function(s) of the plan	Representative image
Local Area Plan (LAP) and Town Planning Scheme (TPS) (Statutory spatial plans) Typical frequency of preparation: 5-20 years (in coordination with the DP/ Masterplan) Prepared by the city planning authority or the Urban Local body e.g., Ahmedabad Urban Development Authority (AUDA), City and Industrial Development Corporation (CIDCO) Approving authority: Municipal Corporation/ Development Authority/ MPC	1:5,000 to 1:1,000 (as per the requirement of the project)	LAP: Rationale for LAP - Existing Plot-level land use data - Detailed land use, urban design and infrastructure provision plan - Detailed strategies for heritage and/or environment preservation - Detailed building regulations TPS: Rationale for TPS - Plot amalgamation and reconstitution strategy and detailed plot level plan - Detailed land use, urban design and infrastructure provision plan - Detailed strategies for heritage and/or environment preservation - Detailed building regulations	A detailed sub-city land use plan along with integration of urban infrastructure, transport and services. LAP is typically prepared for brownfield areas (for heritage conservation/ environmental protection) and TPS for greenfield areas (facilitating planned expansion of urban core).	 e.g., Town Planning Scheme No.2, NAINA, in Navi Mumbai, Maharashtra⁹

⁹ City and Industrial Development Corporation of Maharashtra (CIDCO). (2017). *Town Planning Scheme NAINA No. 2, Navi Mumbai*.

Other types of plans in practice include:

- **Special Purpose Plans** (such as Comprehensive Mobility Plan, City Sanitation Plan, Disaster Management Plan, etc.) which identify the needs of specific areas requiring special plans within the framework of the Development Plan
- **Annual Plans** to translate Development Plan in the context of annual physical & fiscal resource requirements and to monitor plan implementation with performance milestones (Investment Plan, Audit and Monitoring Plan)¹⁰.

For facilitating systematic urban development, adequate land resources which are serviced with the necessary infrastructure and public amenities are required to be made available to cater to the projected urban growth. Such development can be achieved through urban planning.

Some of the initiatives by the Central/ State governments to improve quality of urban planning in the country include:

- Central assistance for preparation of GIS-based Masterplans for 500 AMRUT cities, 2015
- Model Building Bye Laws, 2016
- Guidelines for Transfer of Development Rights (TDR) by NITI Aayog, 2020
- XVth Finance Commission grants for incubation of new cities, 2021
- Preparation of report on Reforms in Urban Planning Capacity in India by NITI Aayog, 2021
- Summary report on Cities as Engines of Growth by NITI Aayog, 2022
- Designation of existing academic institutes as Centres of Excellence for capacity building in urban planning, announced in Budget 2022-23
- High-Level Committee on Urban Planning announced in Budget 2022-23
- Central assistance towards modernisation in bye laws, announced in Budget 2022-23

C. Urban design influences the experiential quality of cities

What is urban design?

Urban design is the creation of city features based on the spatial plans – it focuses on designing the *citizen experience* of public spaces, public infrastructure, transportation systems, landscapes and housing.

While urban planning and urban design have overlapping objectives, they operate on different scales and timelines. While city planning is a precursor for all strategic economic interventions and city-regional scale infrastructural projects, urban design is at a neighbourhood level or local area level. Therefore, both are mutually exclusive and cannot be replaced by each other in training and practice. For example, urban design, focuses on the micro aspects of– for instance, the pedestrian network in the catchment area of a mass transit station, or a public plaza in a heritage area. The focus is on functionality and aesthetics, and the time frame for urban design projects is short-term – say, less than five years.¹¹

It is to be noted here that the city masterplans may not achieve their spatial order objectives without local level planning and urban design. At the same time, minor interventions through urban design may not be able to fix city wide urban planning errors. For example, footpaths may be used as a loop by the pedestrians, nonetheless, providing them in patches might not ensure walkability in the city.

Some of the initiatives by the Central/ State governments to make urban design an integral part of the urban development process include:

- Formulation of **street design guidelines** for multiple cities in India (for e.g., the Urban Street Design Guidelines for Pune guidelines for pedestrian and bicycling elements, and roadside, carriageway and

¹⁰ Ministry of Urban Development. (2015). *Urban and Regional Development Plans Formulation and Implementation Guidelines*. Delhi: Ministry of Urban Development.

¹¹ Urban Design Lab. (2022, October 1). *Difference between Urban Planning and Urban Design*. Retrieved 1 October 2023.

intersection design.¹²

- The **Complete Streets Framework toolkit 2019** (comprising of Complete Streets Policy Framework, Policy Workbook, Planning Workbook, Design Workbook, Implementation Workbook, Evaluation Metrics, and Best Practices) prepared by Institute for Transportation and Development Policy (ITDP) India for the 100 Smart Cities to prioritise walking, cycling, and public transport over cars and unlocking the inherent potential of street space.
- The **Streets for People Challenge**, launched in 2020, is an initiative of the Smart Cities Mission, Ministry of Housing and Urban Affairs (MoHUA), Government of India, to inspire cities to create walking-friendly streets through quick measures in response to COVID-19. (Streets for People Challenge – ITDP India)¹³

What are the key things that Master plans/development plans put forth?

Planning area boundary

A section in the Masterplan/ DPs defines the geographical boundaries of the planning area. It specifies the administrative extent within which the development plan will be applied, including its outer limits and any associated sub-regions.

Population projection:

Based on past trends, the Masterplan/ DP report provides an estimate of the expected population growth within the planning area, which is crucial for infrastructure and service provision planning.

Proposed land use zones:

This spatial plan defines how land within the planning area will be allocated for various purposes such as residential, commercial, industrial, and open spaces.

Capital Projects for next 10-20 years:

It lists major infrastructure and development projects required to support the plan's goals, including transportation, utilities, and public facilities.

Phasing and implementation plans:

This section contains roles and responsibilities of all stakeholders, resource mobilisation framework and phase-wise implementation schedule for planned project activities. The key aspects to be covered in the Implementation framework is intended to align with aims, objectives and prioritised projects and schemes.

Development control regulations or Building Bye Laws:

For the uniform implementation of urban planning and urban design proposals across the planning jurisdiction, development control and promotion regulations are developed by the city/ state governments. These regulate the manner and extent of use of land in various zones through *zoning regulations (FAR/FSI being one of them)*, standards of building construction-maintenance-repairs and rehabilitation through *building byelaws*, and other facets of urban development such as urban design, fire safety, energy efficiency, open spaces and landscaping, etc.

¹² Pune Municipal Corporation. (2016). *Urban Street Design Guidelines: Pune*. Pune Municipal Corporation.

¹³ Press Information Bureau Delhi. (2022, January 18). *MoHUA announces winners of Streets for People and Nurturing Neighbourhoods Challenge*.

What are the mechanisms used to implement spatial plans?

A. Land appropriation

In order to carve out land for public amenities and infrastructure as well as urban expansion, city authorities 'reserve' land parcels in the masterplan/ development plan. The statutory instruments for appropriation of such private land parcels for implementation of masterplan proposals include:

- **Land Acquisition:** As per the URDPFI guidelines, "Land Acquisition" popularly means the legal acquisition of land reserved for a defined public purpose by a government authority from individual landowners, after paying a government-prescribed compensation to cover the losses incurred by landowners from surrendering their land. The land acquisition process can be undertaken by the State or through private initiatives. Presently, land acquisition processes are governed by the Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation, and Resettlement Act, (LARR), 2013.
- **Accommodation Reservation:** AR allows landowners to develop the sites reserved in the masterplan/ DP for public amenities subject to agreeing to entrust and hand over the built-up area of such amenity to the local authority free of all encumbrances and accept full Floor Space Index (FSI)/ Floor Area Ratio (FAR) as compensation in lieu thereof. In some cities, the landowner can choose to develop the amenity on part-portion of the entire plot – and proceed to utilise the compensation FSI on the balance plot. Reservations such as retail markets, dispensaries, etc. can be implemented in this way wherein local authority is not required to acquire the land by incurring expenditure on payment of compensation.
- **Transfer of Development Rights (TDR):** TDR is a market-based planning instrument which virtually separates the development potential of a land parcel from the physical land and allows the owner to use/ sell it elsewhere within the city. TDR is awarded by the planning authorities or urban local bodies in lieu of monetary compensation, for various purposes – for surrender of land for masterplan reservation (for road widening and/or public amenities), heritage conservation, environmental protection, slum redevelopment, and so on.

As per the city-specific building regulations, TDR can be utilised over and above the permissible FSI/ FAR – thus creating a 'TDR market' of sorts. TDR can be strategically utilised to shift the trajectory of built development from the congested core of the city to the sparsely developed areas, by demarcating "sending" and "receiving" zones in the spatial plan.

- **Land Pooling:** Under the Town Planning Scheme model, individual plot owners within the demarcated planning area collectively surrender their land parcels to the government authority/ development agencies in lieu of a serviced plot. This allows for:
 - Proper planning and regularisation of plot sizes,
 - Allocation of land for basic amenities, creation of a well-planned street network, and
 - Generation of finances for development (through levy of betterment charges on the landowners and sale of government-owned plots),

The reconstituted plots are allotted back to the original landowners, with an average reduction of 20-50% in plot area, but with a much greater real estate value on account of improved infrastructure and amenities. The model has been working effectively internationally like Japan, and in India – in the States of Gujarat and Maharashtra – since decades.

- **FAR/ FSI incentives for specific types of development:** Incentives in the form of additional FSI/ FAR are awarded to plot owners or developers for adherence to certain government objectives or for the construction of public purpose buildings and amenities. This is an effective way of engaging the private

sector for the provision of public services, resulting in a win-win situation. The use of such incentive FSI is, however, governed by the development control regulations prepared by the ULB/ State government.

Some of the common purposes for FSI incentives include development of affordable housing, rental housing, cluster development, slum rehabilitation, redevelopment of derelict buildings, attracting high-density development in Transit-Oriented Development (TOD) zones, 'green' buildings (energy-efficient), and so on.

B. Building Permits

The implementation of zoning and development control regulations – and, by extension, the spatial plan – is monitored by the planning authorities (Urban development authorities/ municipal corporations) by way of issuance of building permits. The government can keep checks on development trends and opt to course-correct (by reviewing the spatial plan and regulations) based on their observations. Such data can also inform decisions regarding the city's infrastructure management, disaster management, and safety requirements.

Under the Ease of Doing Business initiative of Central Government, multiple states have enabled their cities to create online portals – generally referred to as **OBPAS (Online Building Plan Approval System)** – and implement a Building Plan Management System (BPMS), with the vision of standardising various building permission processes for ULBs. These portals are capable of analysing the uploaded building plans within minutes and generating comprehensive analytical reports – thus expediting the approval process by eliminating the need for manual scrutiny.

Some noteworthy cases include the Urban Local Bodies of NCT Delhi and Mumbai, who have streamlined their construction approvals process through OBPAS. The Government of Andhra Pradesh has also introduced Development Permission Management System (DPMS), which has integrated all the urban local bodies of the State to a common portal to give building plan approvals online in an expeditious manner. To expedite the adoption of OBPAS, the Ministry of Housing and Urban Affairs published 'The Guidelines for implementing Online Building Plan Approval System' in 2016 for the guidance of 51 million-plus cities, as part of Ease of Doing Business Initiatives.¹⁴

How is urban development induced or effected by government policies in multiple sectors? —

A. Policy framework

The Government of India and state governments periodically release policies and guidelines related to spatial planning and urban development. The National Urban Policy Framework and various state-level policies provide strategic direction for urban development and spatial planning.

Special township policies and initiatives: Four states in India, as of 2012, have implemented Special Integrated Township policies – Maharashtra, Gujarat, Karnataka and Rajasthan. In areas with large tracts of land under private ownership/ control, Integrated Townships are a mechanism that allow governments to transfer the onus of providing planned high-quality development on to the private sector. For instance, under the Special Township Policy for Integrated Development in Maharashtra, developers building private townships on 100 acres of land or more in the state will be allowed to build and look after infrastructure – internal roads, water supply, electricity distribution and local transport – for the first ten years, after which its governance will be handed over to a civic body.¹⁵ The policy also mandates carving out a set percentage of the township area for public amenities, thereby ensuring equitable access.

¹⁴ Ministry of Urban Development. (2016). *Guidelines for implementing 'Online Building Plan Approval System'*. New Delhi: Ministry of Urban Development.

¹⁵ UDD Maharashtra. (2016, December 26). *Regulations for Development of Integrated Township Projects (STP)*. Directorate of Town Planning Maharashtra.

Tourism policies: States like Maharashtra and Gujarat have formulated policies and incentives to argument tourism infrastructure. For instance, the Gujarat Tourism Policy 2021-25 offers capital subsidy for development of Hotels, Theme Parks, Convention Centres, Wellness Resorts and Wayside amenities, as well as upgradation of tourism infrastructure in the vicinity of major destinations, which will have an influence on the spatial development of tourism centres in Gujarat. Maharashtra on the other hand has formulated various policy for specific sub-sectors like Agro Tourism, Adventure Tourism, Caravan Tourism, and Beach Shack Policy, which details the infrastructure requirements based on the nature of tourism.

Industrial policies: Under the National Industrial Corridor Development Programme, the Government of India has envisaged the Delhi Mumbai Industrial Corridor (DMIC) for industrial development along the Western Dedicated Freight Corridor (DFC) between JNPT, Mumbai and Delhi. Under this project, 24 industrial regions and industrial areas of 100-200 sq. kms. each will be developed along this corridor across the states of Uttar Pradesh, Haryana, Rajasthan, Madhya Pradesh, Gujarat, and Maharashtra. This will be further supported by development of multi-modal logistics parks and provision of affordable housing and services for people working in these logistics parks. Similar industrial corridors have been planned between Amritsar-Kolkata, Chennai-Bengaluru, Mumbai-Bengaluru, and Hyderabad-Nagpur.

Railway policies: Indian Railway Stations Development Corporation (IRSDC) is the nodal agency to develop Railway land entrusted to them by the central Ministry of Railways. IRSDC has set out to develop self-sustainable railway stations in the country with high standards of safety, comfort, user-friendly passenger amenities, value-added services and efficiency by adopting state-of-the-art technology, practices and optimum utilisation of resources, supported by robust financial strategies. IRSDC aspires its commercial development to adopt the principles of Transit Oriented Development (TOD) and Green Building Codes. Therefore, IRSDC has published a comprehensive set of 'Codes for Commercial Development on Railway Land' to regulate Layout Planning and Sub-Plot Development within Railway land, based on the Form-Based Codes approach.

The [website](#) of IRSDC contains a set of Manuals and Guidelines for Station Redevelopment Including Commercial Development.

Ports policies: Under the Sagarmala programme, the Government of India has proposed development of Coastal Economic Zones (CEZs), Coastal Economic Units (CEUs), port-led industrialisation, and Smart Industrial Port Cities for industrial development and improving logistics in conjunction with port development. In addition, the development of an SEZ at JNPT, Mumbai and Smart Industrial Port City (SIPC) at Paradip and Kandla are currently under progress.¹⁶

Therefore, although urban development is a mandate of urban development authorities/ ULBs, interventions by multiple other agencies or land owning agencies also creates or impacts urbanisation.

What is the status of spatial planning and design in India?

A. State of plan preparation, implementation and enforcement in urban areas

Macro-level planning

Statutory towns and Census Towns

The Ministry of Housing and Urban Affairs (MoHUA) launched a 100% centrally funded **Sub-scheme on formulation of GIS based Master Plans under AMRUT Mission** for 500 cities in October 2015. This scheme was extended to Class II towns in the year 2021, under AMRUT 2.0, with the objectives to develop standardised base maps, land use maps, master plans, and urban geo-portal using Geographical

¹⁶ Sagarmala Development Company Limited. (n.d.). *Sagarmala Port-led Industrialisation*. Retrieved 1 December 2023.

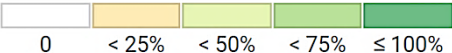
Information System (GIS).

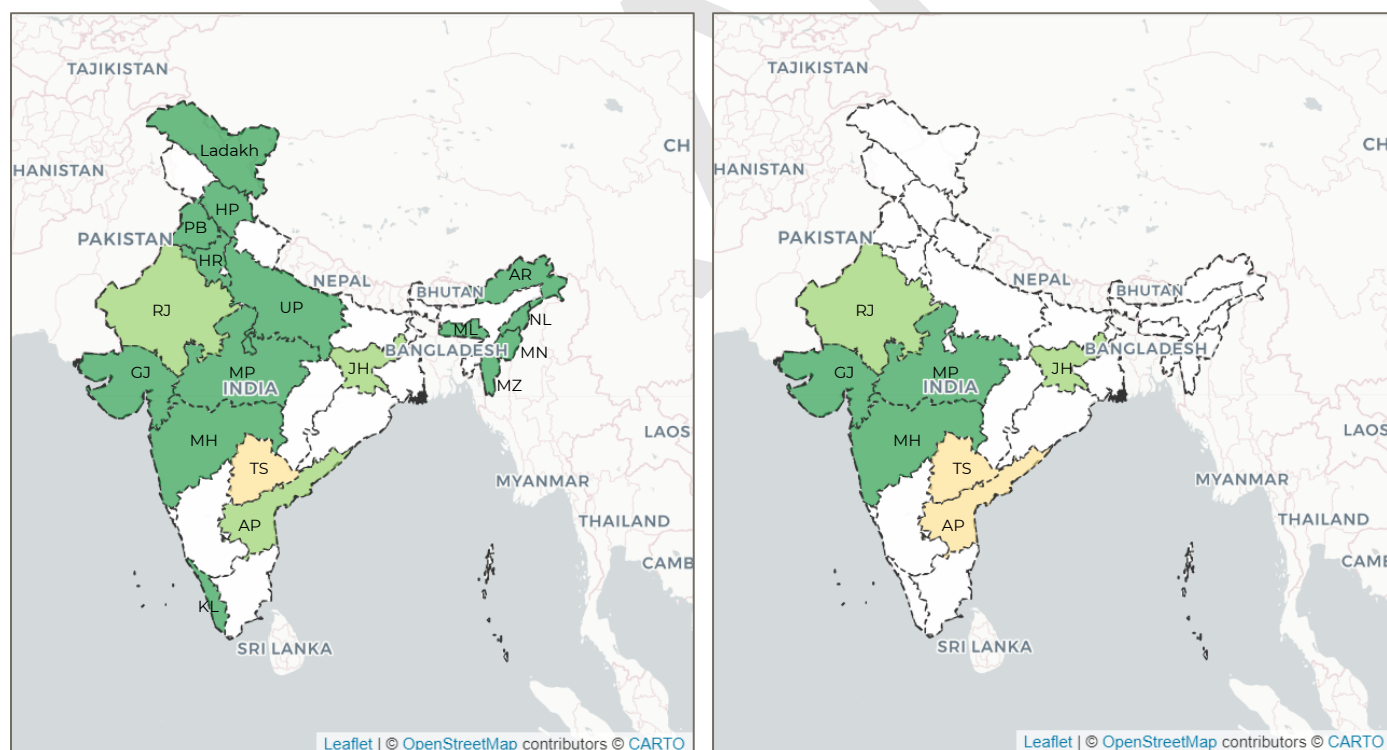
The sub-scheme components¹⁷ are:

- Generation of Base Map and Thematic Maps and *Urban Geodatabase Creation* at the scale of 1:4,000 as per Design & Standards
- Formulation of *Master Plan*: Formulation of Master Plan of city as per State Town & Country Planning Act on the GIS base map and sector-wise data analysis. The implementing agency is State Mission Director/ULBs.
- *Capacity Building*: The training is at three levels:
 - Administrators level – three days duration
 - Planning level – two weeks duration
 - Operators and Technicians level – four weeks duration

About 65% out of the 7933 urban settlements (statutory towns and census towns) did not have any masterplan to guide their spatial and infrastructural development, as of 2021.¹⁸ At present (2023), the status of completion of draft and final Masterplan preparation, as available on the sub-scheme dashboard ([Dashboard link](#)) is as follows:

Figure 3: (Left) Status of draft GIS masterplan preparation for AMRUT cities within the states
Figure 4: (Right) Status of final GIS masterplan preparation for AMRUT cities within the states

Legend: 



Source: TCPO Dashboard, link in above paragraph

Micro-level planning

Status of preparation of Local Area Plans (LAPs) and Town Planning Schemes (TPSs): 80 out of 111 (i.e., 72%) municipalities surveyed under the Municipal Performance Index (MPI) 2020¹⁹, do not implement

¹⁷ Ministry of Housing and Urban Affairs. (2015, October n.d.). *Sub-Scheme on Formulation of GIS Based Master Plans for 500 AMRUT Cities*. Ministry of Housing and Urban Affairs.

¹⁸ NITI Aayog. (2021). *Reforms in Urban Planning Capacity in India*. New Delhi: NITI Aayog.

¹⁹ Ministry of Housing and Urban Affairs. (2021). *Municipal Performance Index 2020*.

master plans through TPSs. 63 of 111 (i.e., 57%) municipalities follow the practice of preparing LAPs.

Figure 5: Status of preparation of micro-level plans for 111 surveyed municipalities



Source: Municipal Performance Index (MPI) 2020¹⁹

Use of land pooling for facilitating implementation of micro-level plans: 78 out of 111 (i.e., 70%) municipalities surveyed under the MPI 2020, do not practice land pooling.

Figure 6: Status of municipalities that practice land pooling for implementing micro-level plans



Source: Municipal Performance Index (MPI) 2020¹⁹

Planning for Disaster and Climate Risks²⁰:

Disaster risk and the impacts of climate change have the potential to disrupt urban development to a massive extent. Thus, it is crucial that city planning accounts for response, resilience and mitigation measures.

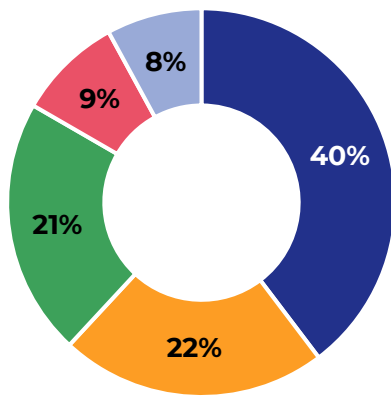
- The Disaster Management Act, 2005, mandates the preparation of **disaster management plans** at the national, state, and district levels. A Disaster Management Plan (DMP), as per the DM Act, is a legally mandated document that outlines strategies and procedures for disaster prevention, preparedness, response, and recovery. It includes resource allocation, coordination among agencies, communication strategies, and legal frameworks to effectively manage natural and man-made disasters. These plans are developed at various administrative levels, such as national, state, and district, to ensure a comprehensive and organised approach to disaster management.
- The National Action Plan on Climate Change (NAPCC) is India's primary policy document outlining its strategy for addressing climate change. Cities are recommended to develop a **climate action plan**, in a participatory manner and in accordance with national guidelines such as Guiding Principles for City Climate Action Planning from UN-HABITAT and the National Mission on Sustainable Habitat, to reduce GHG emissions and reduce the negative impacts of climate change. This would also include a financial plan and allocation of dedicated funds within municipal budget for implementation of the climate action plan.

The ClimateSMART Cities Assessment Framework 2.0 by the Ministry of Housing and Urban Affairs assesses the performance of 126 cities on climate-smart urban planning, rejuvenation and conservation of water bodies, green cover and biodiversity.

CAP: Climate Action Plan

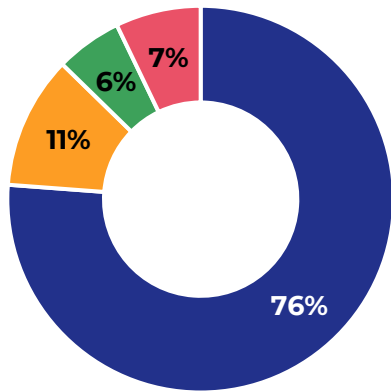
DMP: Disaster Management Plan

²⁰ MoHUA. (2021). *ClimateSmart Cities Assessment Framework 2.0: Cities Readiness Report*. New Delhi: Ministry of Housing and Urban Affairs & Climate Centre for Cities, NIUA.



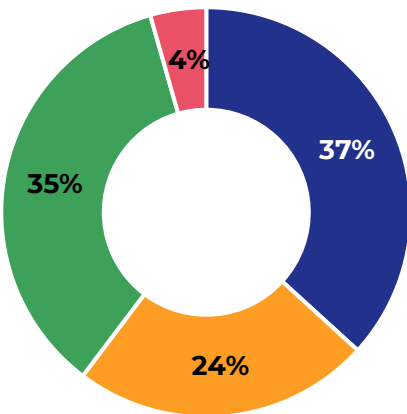
■ Cities do not have a Disaster Management Cell

- Cities have established a Disaster Management Cell
- Cities have developed a DMP
- Cities have established early monitoring systems
- Cities regularly monitor DMP



■ Cities do not have a Climate Action Cell

- Cities have established a Disaster Management Cell
- Cities have initiated the implementation of CAP
- Cities have established early monitoring systems



■ No assessment of water bodies and open areas

- Mapping of water bodies and open areas initiated
- Conservation and rejuvenation measures implemented
- Monitoring and review mechanisms established

How does urban land management relate to ease of doing business?

A. Ease of Doing Business facilitated by Land Management reforms

Urban land reform through land titling laws

At present, India follows a presumptive land titling system, whereby the ownership is established on the basis of current possession.

Clear land titling is a crucial step towards enabling access to land and property under the ambit of law, and is a precursor to sustainable urban planning. It helps reduce land-related litigations, making acquisition easier and real estate transactions fairer and more transparent – thereby speeding up implementation of urban development projects.

82% of the 111 municipalities surveyed under MPI 2020 did not have land titling laws.

Figure 7: Municipalities with land titling laws



Source: Municipal Performance Index (MPI) 2020¹⁹

Provision of Single-window Clearance for granting Building Approvals/Permissions

62% of the 111 municipalities surveyed under MPI 2020 have implemented single-window clearance systems. This is a positive sign for ease of doing business.

Figure 8: Municipalities with Single Window Clearance for Building Permissions



Source: Municipal Performance Index (MPI) 2020¹⁹

Inefficiencies and delays in urban land development processes can be effectively countered by having a one-stop platform for applying for online building plan approvals. Such platforms reduce the time involved in processing building permissions, make the entire process transparent and curb malpractices. Under EoDB, (mention indicator on time taken for building approval).

Cities like Mumbai and states of Tamil Nadu and West Bengal have successfully implemented and operationalised single-window portals for obtaining building approvals.

How is the spatial planning and design sector governed in India?

A. Spatial Planning – an Urban Local Body subject

The **74th Constitution (Amendment) Act, 1992** provides a constitutional status to Urban Local Bodies (ULBs) as the third tier of government. Furthermore, the 12th Schedule of the Constitution lists functions specific to spatial planning and design, which may be devolved to ULBs by the states. These functions are as follows:

- Function 1: Urban planning, including town planning
- Function 2: Regulation of land use and construction of buildings
- Function 12: Provision of urban amenities and facilities such as parks, gardens and playgrounds
- Function 17a: Public amenities including street lighting, parking lots, bus stops, and public conveniences

Allied functions which would be directly influenced by spatial planning and design include:

- Function 4: Roads and bridges
- Function 10: Slum improvement and upgradation

B. State of devolution of powers to Urban Local Bodies

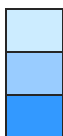
While the 74th CAA recommends devolution of powers to ULBs, 17 states have partially devolved and only

one state (Mizoram) has completely devolved the urban planning function to ULBs. Devolution of urban design powers is relatively more successful.

Table 2: Devolution of functions specific to spatial planning and design out of the 18 functions in 74th CAA (extracted in part from the Praja Urban Governance Index)

Legend:

Functions under city government
Functions under multiple agencies
Functions under State government



States	1: Urban planning including town planning	2: Regulation of land-use and construction of buildings	12: Provision of urban amenities and facilities such as parks, gardens, playgrounds	17a: Public amenities including street lighting, parking lots and public conveniences
Andhra Pradesh	Functions under city government	Functions under city government	Functions under city government	Functions under city government
Arunachal Pradesh	Functions under State government	Functions under State government	Functions under city government	Functions under multiple agencies
Assam	Functions under State government	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies
Bihar	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies
Chhattisgarh	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies	Functions under city government
Delhi	Functions under multiple agencies	Functions under State government	Functions under multiple agencies	Functions under multiple agencies
Goa	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies
Gujarat	Functions under State government	Functions under State government	Functions under multiple agencies	Functions under city government
Haryana	Functions under State government	Functions under State government	Functions under city government	Functions under multiple agencies
Himachal Pradesh	Functions under State government	Functions under multiple agencies	Functions under city government	Functions under city government
Jharkhand	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies	Functions under city government
Karnataka	Functions under multiple agencies	Functions under multiple agencies	Functions under city government	Functions under multiple agencies
Kerala	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies	Functions under city government
Madhya Pradesh	Functions under multiple agencies	Functions under multiple agencies	Functions under city government	Functions under multiple agencies
Maharashtra	Functions under multiple agencies	Functions under multiple agencies	Functions under city government	Functions under city government
Manipur	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies
Meghalaya	Functions under State government	Functions under State government	Functions under multiple agencies	Functions under multiple agencies
Mizoram	Functions under city government	Functions under city government	Functions under multiple agencies	Functions under multiple agencies
Nagaland	Functions under State government	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies
Odisha	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies	Functions under city government
Punjab	Functions under State government	Functions under State government	Functions under multiple agencies	Functions under multiple agencies
Rajasthan	Functions under multiple agencies	Functions under State government	Functions under multiple agencies	Functions under multiple agencies
Sikkim	Functions under State government	Functions under multiple agencies	Functions under multiple agencies	Functions under multiple agencies

States	1: Urban planning including town planning	2: Regulation of land-use and construction of buildings	12: Provision of urban amenities and facilities such as parks, gardens, playgrounds	17a: Public amenities including street lighting, parking lots and public conveniences
Tamil Nadu				
Telangana				
Tripura				
Uttar Pradesh				
Uttarakhand				
West Bengal				

Source: Praja Foundation²¹

C. Legislative framework for spatial planning

In India, urban planning processes and preparation of masterplans/ development plans are governed by the states' respective planning legislations/ frameworks.

- The **Constitution of India** provides a framework for the division of powers between the central and state governments. While **land and urban planning primarily fall under the jurisdiction of state governments**, certain aspects related to spatial planning, such as protection of the environment and inter-state river water disputes, may involve the central government.
- The respective **State Municipal Acts** govern the constitution, powers and responsibilities of urban local bodies. *For e.g., the Maharashtra Municipal Corporations Act, 1949*
- The **State Town and Country Planning Acts** provide the legal basis for urban and regional planning within the respective states. They outline the roles and responsibilities of planning authorities, development controls, zoning regulations, and land use policies. *For e.g., the Maharashtra Town and Country Planning Act, 1966*
- The **Development Authority Acts** facilitate establishment of an authority to plan, coordinate and supervise development at the regional level (essentially urban centres and their agglomeration areas). *For e.g., The Mumbai Metropolitan Region Development Authority Act, 1974*
- The **Special Area Development Authority Acts** facilitate establishment of an authority for planning, development and administration of special areas notified by the State Government. *For e.g., the Maharashtra Housing and Area Development Act, 1976 governs the laws related to housing, repair and reconstruction of dangerous buildings and improvement of slum areas in the state.*
- The **Urban Improvement Trust Acts** provide for the establishment of Improvement Boards for the development of urban areas in certain states like Karnataka, Rajasthan, Odisha, Punjab, Maharashtra, etc. *For e.g., the Rajasthan Urban Improvement Act, 1959 or the Bombay Improvement Act, 1898 (which established the Bombay City Improvement Trust)*
- **Land Acquisition Laws:** Land acquisition and rehabilitation laws are crucial for spatial planning, as they govern the process of acquiring land for public infrastructure projects, urban expansion, and other development purposes. The LARR Act of 2013 and state-specific legislation address land acquisition and compensation issues.
- Other land appropriation instruments (such as TDR, AR, land pooling) are generally legalised by way of

²¹ Praja Foundation. (2020). *Praja's Urban Governance Index 2020*.

the Town and Country Planning Acts of the respective states.

- **Environmental Laws:** Environmental regulations and laws, such as the Environment (Protection) Act, 1986, and the Water (Prevention and Control of Pollution) Act, 1974, play a significant role in spatial planning by regulating land use, industrial activities, and infrastructure projects to protect the environment.
- **Building Byelaws and Development Control and Promotion Regulations:** Each state in India and the major cities have formulated their own development control regulations that govern construction, building permits, and development standards within urban areas. These rules ensure that development is conducted in a planned and regulated manner.
- **Disaster Management Laws:** The National and State-level Disaster Management legislation is relevant to urban planning as it mandates disaster risk assessment, mitigation, and preparedness, influencing land use, infrastructure, and design decisions to create safer and more resilient urban environments.
- The legal framework for disaster management is primarily governed by the Disaster Management Act, 2005. This comprehensive legislation was enacted to provide for effective management of disasters, including natural and man-made disasters, and to establish institutional mechanisms for disaster management at the national, state, and district levels. The Act mandates the preparation of disaster management plans at the national, state, and district levels. These plans outline the strategies, procedures, and guidelines for various phases of disaster management, including prevention, mitigation, preparedness, response, and recovery.
- **Land Ceiling Laws:** Some states have land ceiling laws that restrict the maximum amount of land an individual or entity can own. These laws are aimed at preventing land concentration and promoting equitable land distribution. The Urban Land (Ceiling and Regulation) Act, 1976 is still in force in the States of Andhra Pradesh, Assam, Bihar, Orissa, Maharashtra and West Bengal despite the Government of India having already repealed the same with effect from January 11, 1999 (.
- **The Transfer of Property Act, 1882,** governs property transfers, outlines various methods of property transfer, sets conditions for valid transfers, and covers important aspects of property transactions, such as mortgages, leases, and licenses. The Act was amended in 2002, with the aim to modernise and simplify property transactions, making them more adaptable to evolving societal and economic needs while providing legal clarity for various property dealings.
- **The *Benami* Transactions (Prohibition) Act, 2016,** supersedes the older *Benami* Transactions (Prohibition) Act of 1988. This newer Act provides a more robust framework for identifying and dealing with *benami* transactions, where property is held in the name of someone other than the actual owner. It empowers authorities to confiscate and manage such *benami* properties, imposing penalties on those involved in these transactions. The 2016 Act also establishes designated authorities for adjudicating *benami* disputes and enhances the government's ability to combat tax evasion and black money, thus playing a crucial role in promoting transparency, accountability, and combating financial malpractice in India.
- **Heritage and Conservation Laws:** Legislation related to heritage conservation and preservation of historic buildings and sites plays a role in spatial planning, particularly in older cities and heritage areas. Some of these laws include the Ancient Monuments and Archaeological Sites and Remains Act, 1958, and the National Monuments Authority (NMA) Act, 2010.

D. Codes and guidelines

- **Urban And Regional Development Plans Formulation And Implementation (URDPFI) Guidelines, 2015:** The URDPFI guidelines were developed by the Ministry of Urban Development (MoUD), Government of India, to provide a framework for the preparation and implementation of urban and regional development plans in the country. These guidelines aim to promote sustainable and balanced urban development, improve the quality of life in urban areas, and guide local authorities in planning

and development activities.

- **National Building Code of India, 2016:** It is a comprehensive building code that provides guidelines and regulations for the design, construction, and maintenance of buildings and infrastructure in India. It is intended to ensure the safety, health, and sustainability of built environments. It is a vital tool for architects, engineers, builders, and other professionals involved in the construction industry.
- **Model Building Bye Laws (MBBL), 2016:** In 2004, the Ministry of Urban Development published the Model Building Bye Laws, in view of Bhuj Earthquake that occurred in 2001, to lay focus on structural safety of buildings and for the guidance of the State Governments. This was subsequently revised in 2016 to incorporate salient features such as barrier-free environment, adoption of modern construction technology, Ease of Doing Business, Swachh Bharat Mission, rainwater harvesting, and effects of communication technology.
- **State Land Revenue Codes, (mention year):** Land revenue codes in India are legal frameworks and statutes that govern the assessment and collection of land revenue, as well as the management and administration of land records. These codes vary from state to state, as land is a state subject under the Constitution of India, and each state has its own land revenue laws.
- **Reforms in Urban Planning Capacity in India, 2021:** This publication by NITI Aayog discusses the key urban challenges in India and the immense relevance of 'planning' skills to address these challenges. It recommends reforms in Town & Country Planning Acts, Institutional capacity building, enhancing the role of private sector, citizen engagement and strengthening of the urban planning education ecosystem.
- **ClimateSMART framework 2.0 (2020):** The Climatesmart Cities Assessment Framework is a toolkit to assess the climate mitigation and assessment framework across different sectors in the Indian urban context. These sectors include Energy and Green Buildings, Urban Planning, Biodiversity and Green Cover, Mobility and Air, Water Management and Waste Management.
- **Smart Cities Mission Guidelines (2015):** The Smart Cities Mission website provides a guidance document for cities participating in the Smart Cities Mission, which aims to promote sustainable and technology-driven urban development.
- **AMRUT 2.0 Reforms Toolkit (Atal Mission for Rejuvenation and Urban Transformation) (2021):** This toolkit assists cities and towns in implementing AMRUT projects related to water supply, sanitation, and urban infrastructure.
- **Environmental Impact Assessment (EIA) Guidance Manuals (2010):** EIA guidelines are essential for assessing the environmental impact of development projects, ensuring that environmental considerations are integrated into urban planning. PARIVESH, under Ministry of Environment, Forest and Climate Change, published a set of EIA guidance manuals in 2010 pertaining to various activities including townships and area development, building construction, highways, airport, mining, etc.
- **Heritage Conservation Guidelines:** Cities with historic areas often have specific heritage conservation guidelines to protect and preserve culturally significant buildings and sites. Some of these include Handbook of Conservation of Heritage Buildings (2013) by the Central Public Works Department, and Model Heritage Regulations (2011) by the Ministry of Urban Development. In many cases, these guidelines are included in the respective city's Development Control Regulations.
- **National Disaster Management Guidelines:** These guidelines help cities and urban areas plan for disaster management and preparedness, including mitigation and response strategies. The National Disaster Management Authority periodically publishes guidelines covering specific types of disasters such as earthquakes, floods, cyclones, drought, etc. on the official website.
- **Street Design Manuals:** Some cities (Pune, Delhi, etc.) and states have street design manuals or guidelines that promote pedestrian-friendly streets, safe road designs, and urban beautification.
- **Solid Waste Management Manuals:** As part of the Swachh Bharat Mission, a two-volume Municipal

Solid Waste Management Manual was released by the Ministry of Urban Development in 2016. These manuals provide guidance on the management and disposal of solid waste in urban areas, including waste segregation and recycling.

E. Institutions involved in managing spatial planning and design

Ministry of Housing and Urban Affairs (MoHUA)	The central ministry responsible for formulating national policies and programs related to urban development and housing. It oversees flagship urban development missions like the Smart Cities Mission, AMRUT and PMAY, and provides financial support, guidelines, and technical assistance to states and union territories for urban development projects.
Town and Country Planning Organisation	TCPO is responsible for providing planning guidance at the national level. It coordinates urban and regional planning activities across states and provides expertise in town planning, land use planning, and development regulations. TCPO plays a key role in shaping national planning policies.
Urban Development Corporations	Urban development corporations provide financial assistance and support for urban infrastructure and housing projects. They play a vital role in urban development financing by extending loans and funding for public infrastructure projects, ensuring that projects adhere to sustainable and urban planning principles. (e.g., HUDCO)
Ministry of Environment, Forest and Climate Change	Responsible for all activities related to protection of environment and forests, and planning, coordination and implementation of any policy, programme or mission related to environment and climate change
Urban Development department OR State Town and Country Planning departments	Each state in India has its own town planning department responsible for urban and regional planning within the state. State T&CP departments formulate masterplans, building regulations, and development control rules for cities and towns in the respective states. They play a critical role in guiding and regulating local urban development.
District Planning Committees (DPC) and Metropolitan Planning Committees (MPC)	DPCs are responsible for formulating District Development Plans (DDPs) that integrate rural and urban development priorities within a district. MPCs are responsible for preparing Regional Development Plans (RDPs) that encompass land use, transportation, infrastructure, and environmental sustainability. Both DPCs and MPCs oversee resource allocation and monitor development projects to ensure well-coordinated growth.
Urban Development Authorities	Development authorities are specialised bodies responsible for planned development in specific metropolitan or regional areas. They plan and execute large-scale urban development projects, including infrastructure, transportation, housing and commercial developments.
Urban Local Bodies	Municipal corporations, Municipalities etc. are responsible for urban governance and development at the local level. They prepare and implement

	masterplans, issue building permits, provide basic services like water supply and sanitation, and manage local infrastructure. ULBs are responsible for day-to-day urban administration and development.
City Improvement Trusts	Organisations that focus on urban development and improvement within specific cities or urban areas. Their role includes urban planning, infrastructure development, housing projects, and the implementation of initiatives aimed at enhancing the quality of life in urban areas.
Industrial development authorities/Special Purpose vehicles	Government entities or agencies responsible for promoting industrial growth, infrastructure development, and investment within designated regions e.g., Gujarat International Finance Tec-City (GIFT), Dholera Special Investment Region (SIR) and AURIC (Aurangabad Industrial City)

What are the central government's priorities?—

The **Atal Mission for Rejuvenation and Urban Transformation (AMRUT), 2015**, primarily focused on ensuring that cities have access to basic infrastructure and services related to water supply, sewerage, drainage, urban transport, and green spaces.

Under Atal Mission for Rejuvenation and Urban Transformation (AMRUT), a sub-scheme on 'Formulation of GIS based Master Plan for AMRUT Cities' has been approved to facilitate States and cities to take up Master Planning and is being implemented in 456 towns of 35 States/ UTs. Further in 2021, under AMRUT 2.0, the sub-scheme was extended to Class II towns as well (*details in the section '[What is the status of spatial planning and design in India?](#)'*).

Central Finance Commission (2021 to 2026)

The CFC recommended allocation of INR 1,21,055 crore for ULBs for the period 2021 to 2026. (XV Finance Commission, 2020). The CFC has also recommended INR 8,000 crore for incubation of new cities and INR 450 crore for shared municipal services.

Entry level conditions recommended by the CFC for accessing grants are related to disclosure of municipal accounts, setting of floor rates for property tax and increase in municipal revenue commensurate to the growth of Gross State Domestic Product (GSDP).

Central Government Budget for 2022-23 and 2023-24

2022-23: The budget speech of the finance minister for 2022-23 highlighted the following points related to urban planning:

- Designation of up to five existing academic institutes as Centres of excellence – with endowment funding of Rs. 250 crore – for developing India-specific knowledge in urban planning and design, and to deliver certified training in these areas
- INR 6,000 crore allocated to Urban Reforms (Part-VI) under Scheme of Special Assistance for Capital Investment 2022-23 for the reform of Modernization of Building Byelaws, Adoption of Transferrable Development Rights (TDR), Implementation of Local Area Plans (LAP) and Town Planning Schemes (TPS), Implementation of Transit-oriented Development (ToD), Creation of Sponge Cities, Removing Taxation for running the Buses for Public Transport. Funds amounting to 4,093 crore have been released by the Department of Expenditure, Ministry of Finance, to 12 States.²²

2023-24: The budget speech of the finance minister for 2023-24 highlighted the following points related

²² Ministry of Finance. (2022, February). *Budget at a Glance 2022-2023*. Government of India. New Delhi.

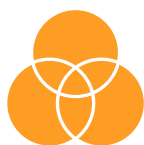
to urban development.

- Encouraging states and cities to undertake urban planning reforms
- Improving credit worthiness of cities to raise municipal bonds
- Setting up of an Urban Infrastructure Development Fund (UIDF) to finance urban infrastructure in Tier-2 and Tier-3 cities
- INR 15,000 crore earmarked for Urban Reforms (Part-III) under Scheme of Special Assistance for Capital Investment 2023-24. The reform components include Augmentation of human resources for strengthening urban planning ecosystem, Implementation of Town Planning Scheme (TPS)/ Land Pooling Scheme, Modernization of Building Bylaws, promoting in-situ slum rehabilitation, Transit-Oriented Development (TOD), Transferable Development Rights as planning tool, Strengthening natural ecosystems of urban areas through urban planning.²³

²³ Ministry of Finance. (2023, February). *Budget at a Glance 2023-2024*. Government of India. New Delhi.

What are potential areas of intervention going forward?

At the National Level



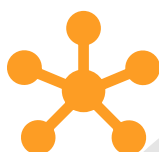
Integration of economic and mobility strategies with spatial planning

While it is generally not mandated by the state planning acts, Masterplans/ DPs can include a section outlining the economic goals and objectives of the plan, including strategies for economic growth, employment generation, and sustainability. This is useful for tying together the spatial and economic growth objectives of the urban area.

The current spatial planning framework is disconnected from the economic aspirations of urban areas due to lack of mandate and multiplicity of authorities playing role in economic development that induces urban growth. Data on economic activity and housing affordability can guide the preparation of realistic city masterplans. However, such economic data for city regions is scattered and often not from same time period. Additionally, mobility plans today are restricted in their ability to influence spatial development.

Institutional frameworks that allow for preparation of a comprehensive urban agglomerations level plans which integrates economic development, land use and transportation strategies may be explored.

At the State Level



Preparation of district level spatial strategies/regional plans

The development of urban centres cannot occur in silos. It is important to foster regional linkages to facilitate future growth.

The urban planning framework can make provision for the preparation of district level spatial strategies, taking into account development trends in the regional context of the district and proposing spatial strategies to be integrated in the spatial plans within the district.

City-regions can adopt a growth driver-based approach, whereby an area's unique identity and strengths are used as a competitive advantage for creating and sustaining growth, and involves policies tailored to regional characteristics



Identifying and measuring key performance indicators for masterplans

At present, there is no mechanism for monitoring the outcomes of spatial plans. Thus, urban challenges remain unaddressed and spatial growth tends to violate the masterplan vision.

While spatial planning and design legislation and policies are said to be based on multiple considerations, including those on climate change, gender and socio-economics, there is no mechanism in place to monitor if these goals are achieved on ground.

Spatial plans can be accompanied by a set of measurable outcomes to be linked with the plans. These outcomes shall pertain to the economic and climate risk management goals of the region, as well as gender and social inclusion. *For instance, outcome indicators of good climate risk*

management could include the presence of a climate action cell, the assessment of water bodies and open areas, budgetary allocation for adaptation and mitigation projects, etc.



Improving masterplan implementation rate

Masterplans/ Development plans tend to remain on paper due to lack of strategy for implementation, lack of capital investment plan, specific roles assigned to various authorities, shortfall of financial resources and capacities for project structuring and implementation also affect the motivation as well as capability of the ULBs to undertake masterplan implementation.

The masterplan report should outline an implementation roadmap for the plan.

A prioritisation process can also inform the budgetary allocation in order to achieve the masterplan objectives in the determined timeline.

City authorities can include land value capture financing strategies as part of their spatial planning proposal, as this could generate adequate financing for upfront investment on masterplan implementation.

Institutional and financial capacity building should be undertaken for micro-level planning.

Capital investment plans (not mandated by Acts but useful): This section, while not mandatory, helps prepare a more realistic and implementable masterplan/ DP, as it details the financial aspects of the plan, including funding sources, budget allocation, and cost estimates for proposed projects.

Carrying capacity analysis (not mandated by Acts but useful): While not mandatory, masterplans/ DPs can include a Carrying capacity analysis which assesses the capacity of the planning area to sustain and support various forms of development, such as population growth, infrastructure, and natural resource use. It considers factors like water availability, environmental impact, transportation, and social services to determine the area's capacity to accommodate proposed development while maintaining sustainability and quality of life. This analysis is vital for ensuring that the development plan aligns with the ecological and resource limitations of the region.



Incorporation of urban design guidelines into development control regulations

The preparation of Form Based Codes (FBC) which integrate urban design guidelines with development regulations can help create contextually appropriate urban forms. Integrating FBCs with the spatial planning framework can help ensure adherence.

FBCs should be required to follow universal accessibility norms, and should be sensitive to the way women, children, differently abled, and the elderly experience urban spaces.

Recommended Readings

- Ministry of Housing and Urban Affairs. (2017). *National Transit Oriented Development (TOD) Policy*. https://mohua.gov.in/upload/whatsnew/59a4070e85256Transit_Oriented_Developoment_Policy.pdf
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